

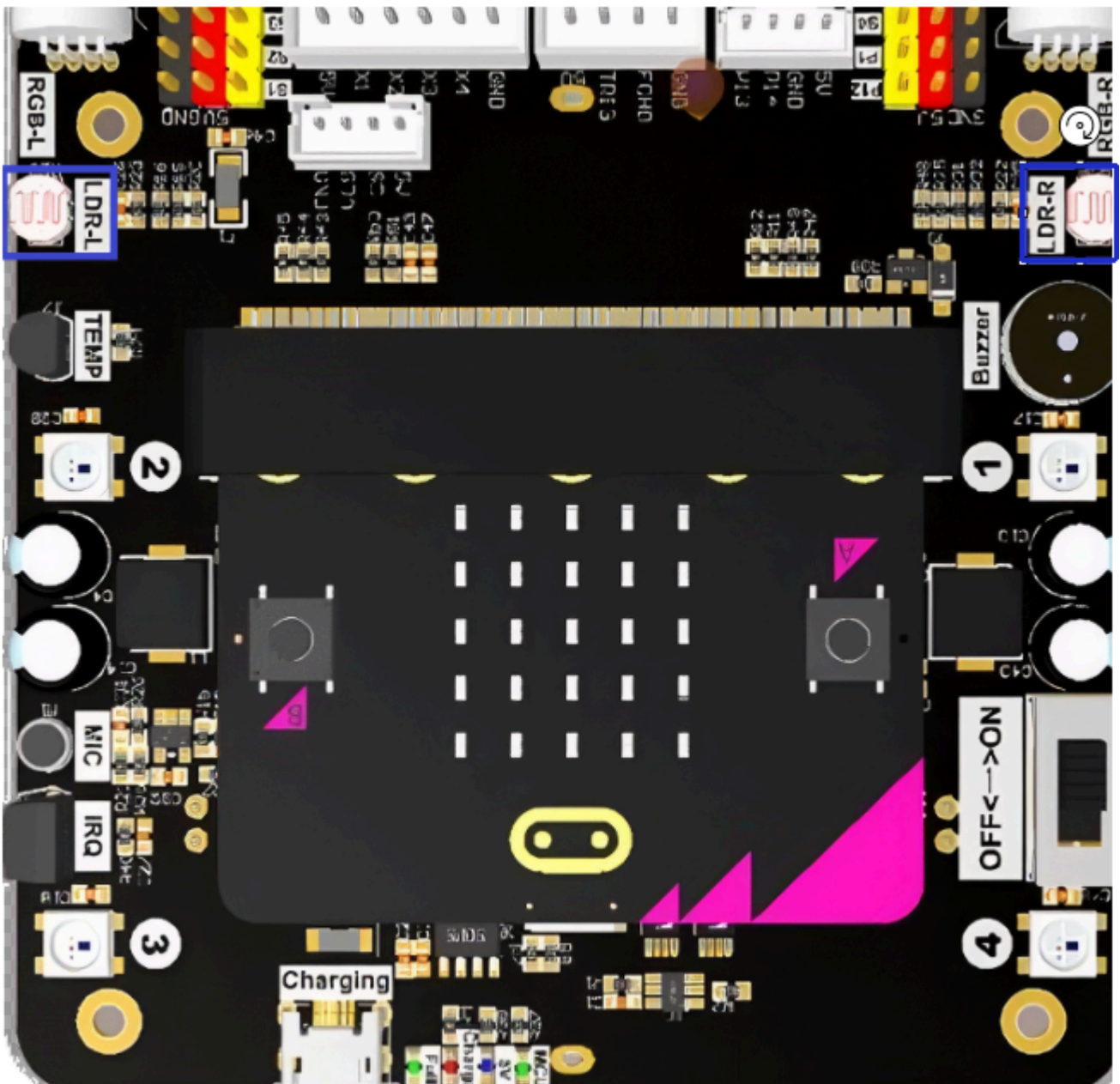
Photosensitive Sensor Data Collection

Photosensitive Sensor Data Collection

1. Experiment Purpose
2. Experiment Steps
3. Experiment Source Code
4. Program Download
5. Experiment Results

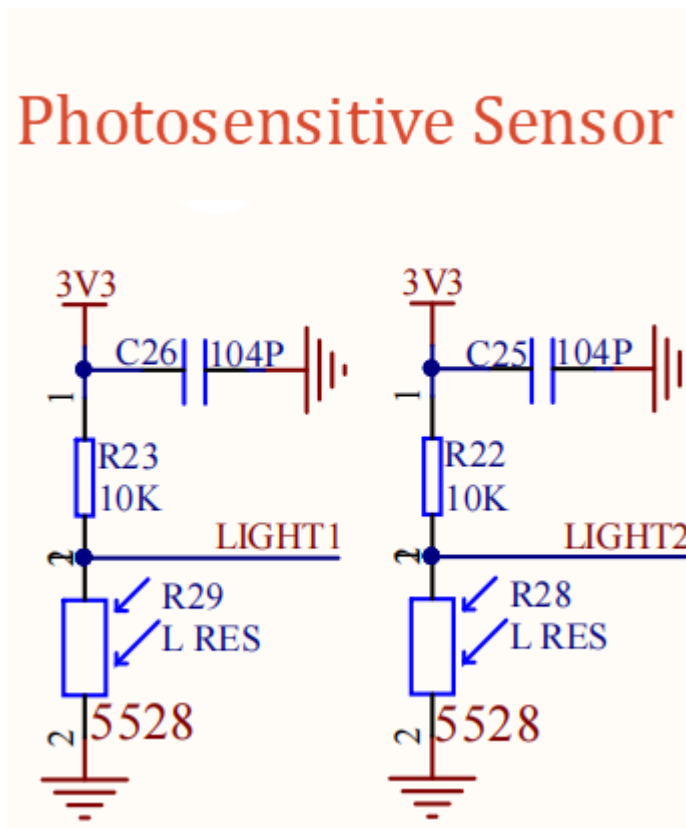
1. Experiment Purpose

The Lastbit car is equipped with two photosensitive sensors on the left and right sides, named LDR-L and LDR-R respectively. This lesson will guide you through learning how to read data from the photosensitive sensors.



Common materials for manufacturing photosensitive sensors include cadmium sulfide, as well as selenium, aluminum sulfide, lead sulfide, and bismuth sulfide. Its working principle is based on the internal photoelectric effect. The stronger the light, the lower the resistance value. As the light intensity increases, the resistance value decreases rapidly. When the light is bright, the resistance value can drop below 1KΩ. Photosensitive resistors are very sensitive to light. In the absence of light, they exhibit a high resistance state. In dim light, the resistance can usually reach 1.5MΩ. Because of its non-contact, fast response, and reliable performance characteristics, photosensitive sensors play a very important role in automatic control and non-electrical electronic technology.

The photosensitive sensor uses a photoresistor voltage-divider light detection circuit to convert the intensity of ambient light into an electrical signal that the controller can read.



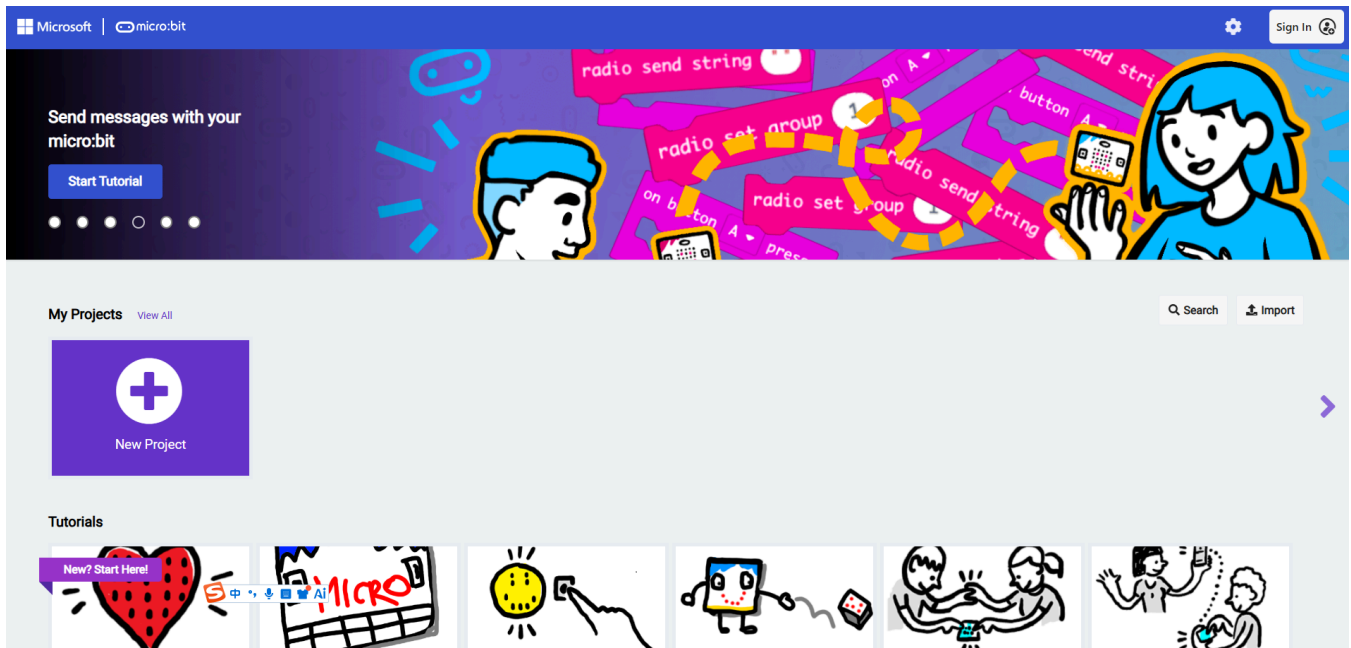
2. Experiment Steps

1. Before starting the experiment, connect the Micro:bit mainboard to your computer using a Micro USB data cable. At this time, the computer will display the MICROBIT drive letter.



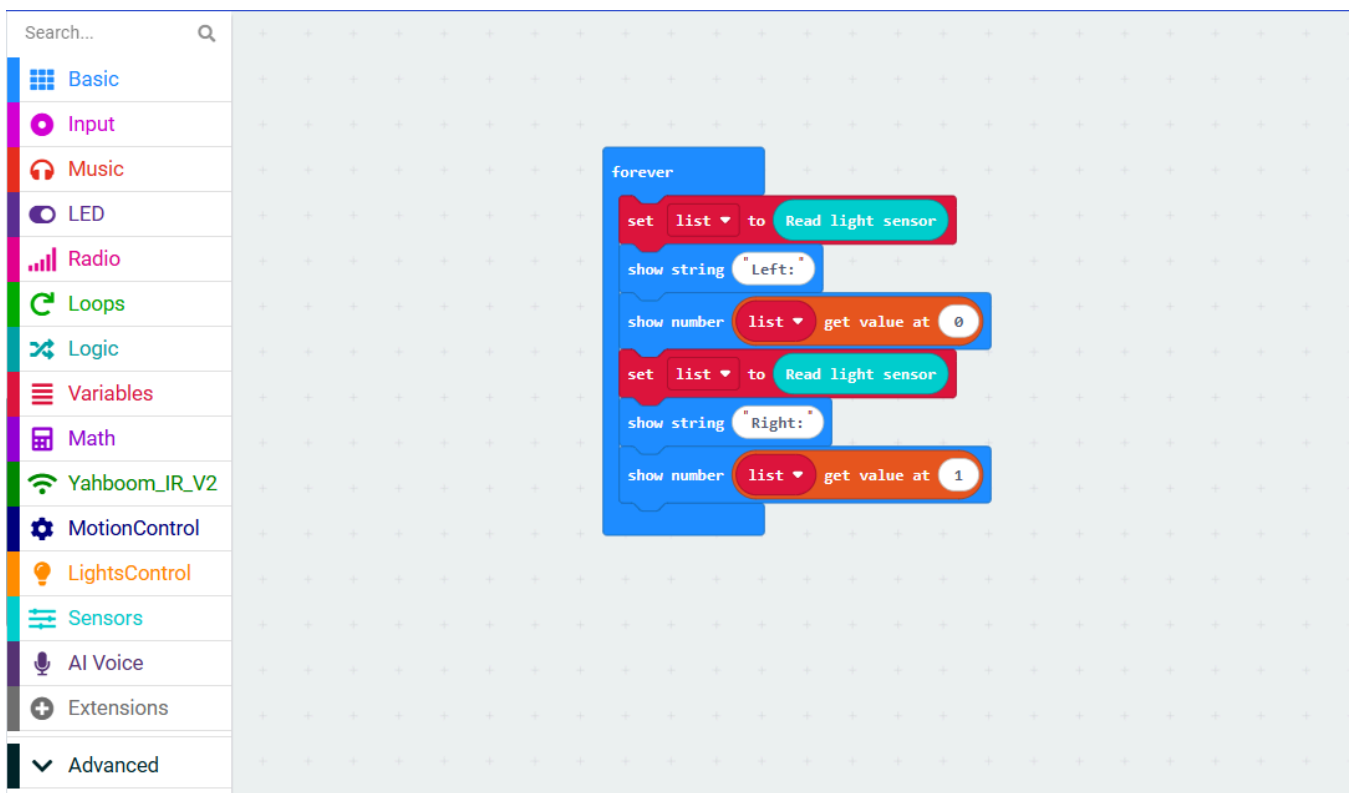
2. Open MakeCode using the link below.

[Microsoft MakeCode for micro:bit](#)

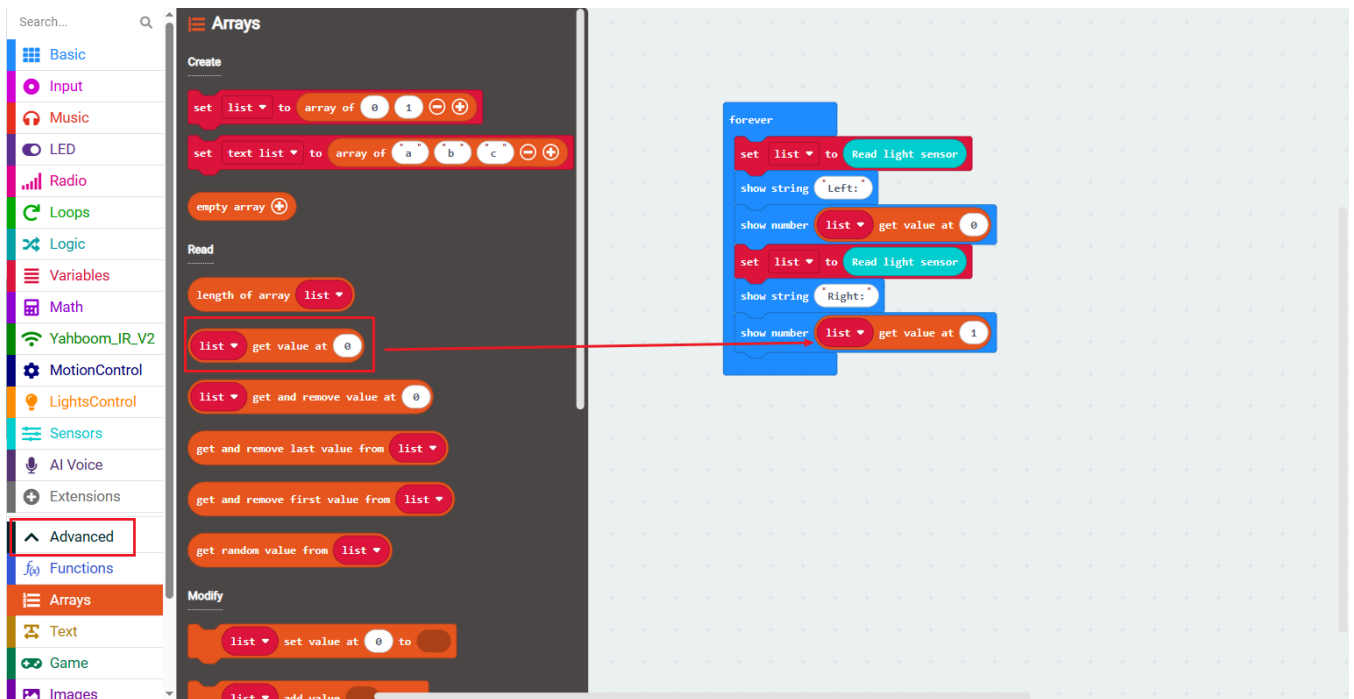


3. Drag `2.Photosensitive_Sensor_Data_Collection.hex` to the new project page. MakeCode will automatically import and open the project.

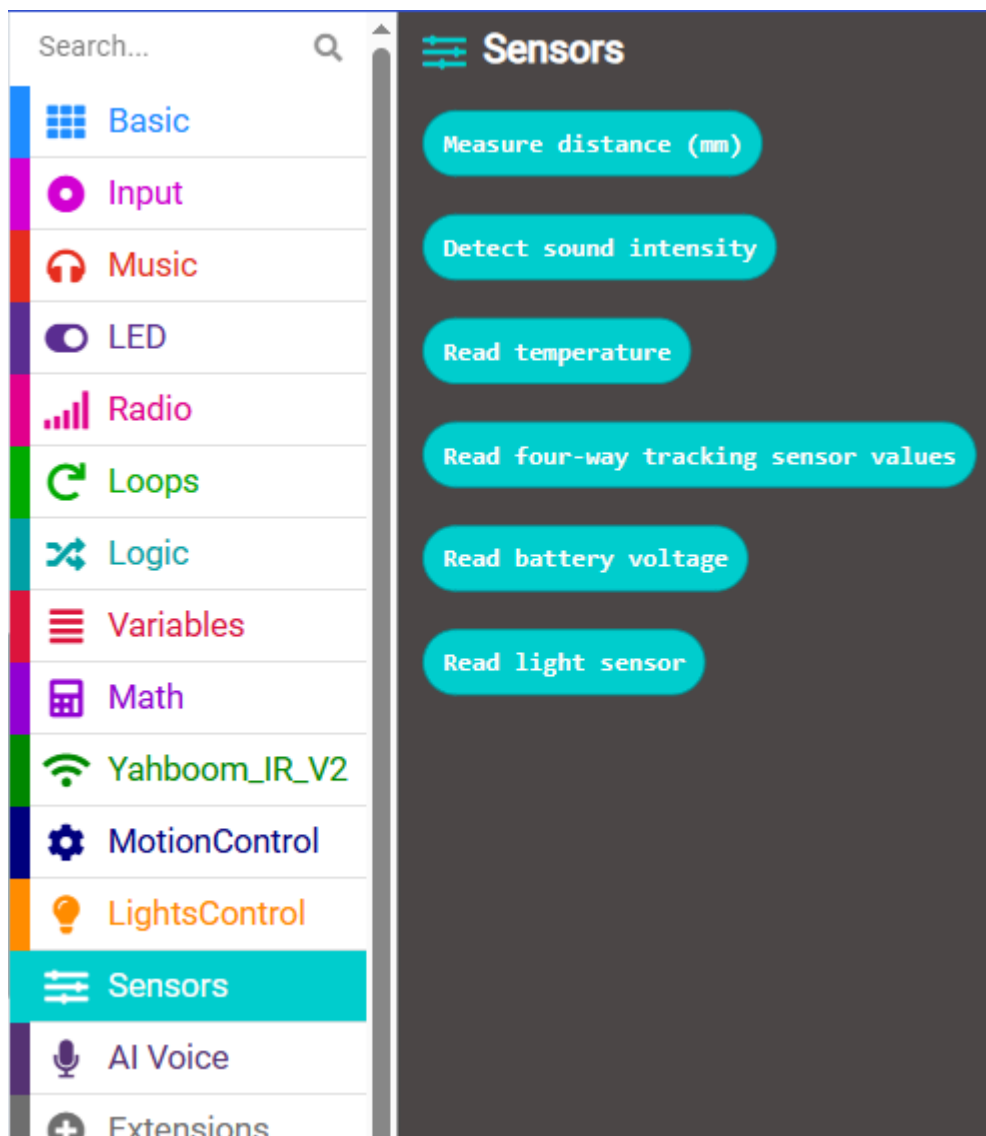
3. Experiment Source Code



The block to get the index value from the array is located in the `Arrays` subcategory under the `Advanced` category.

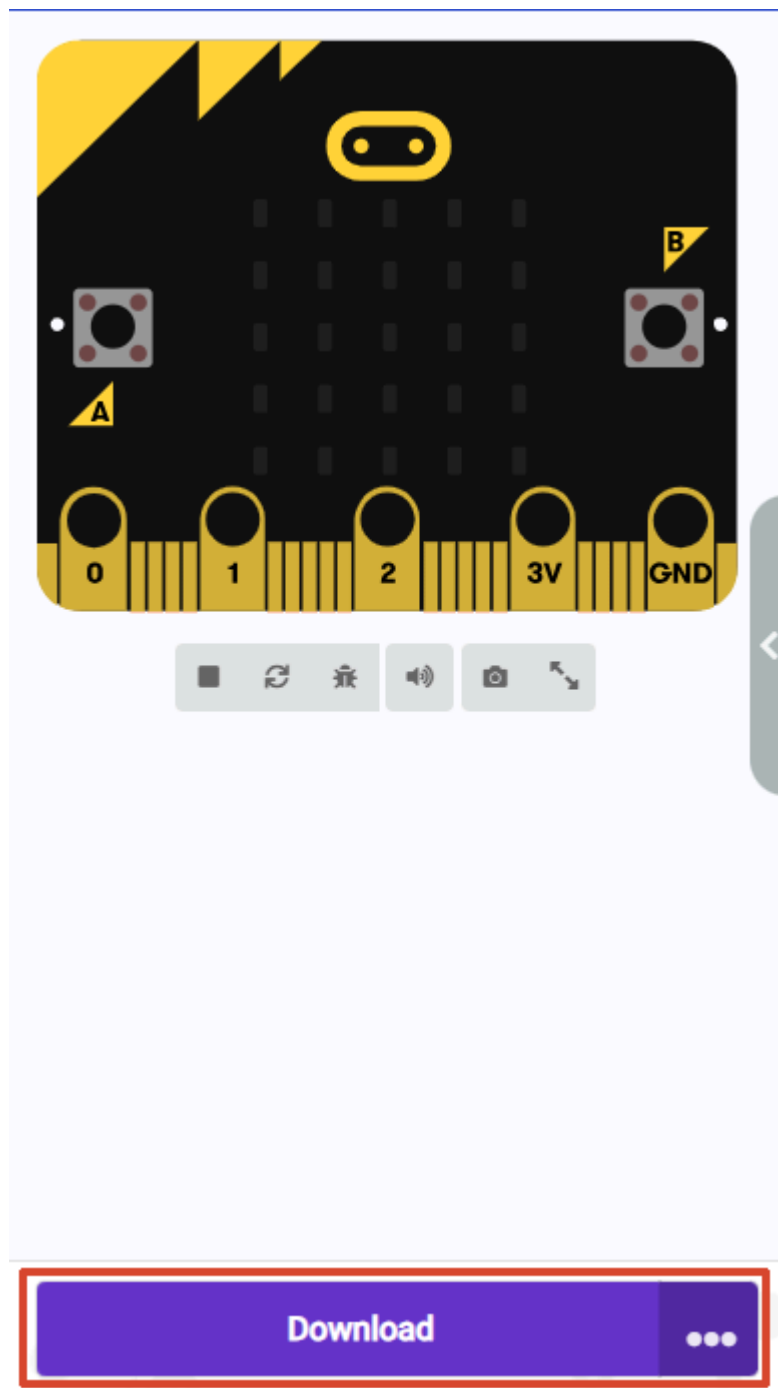


The photosensitive sensor data reading block is located in the **Sensors** category.



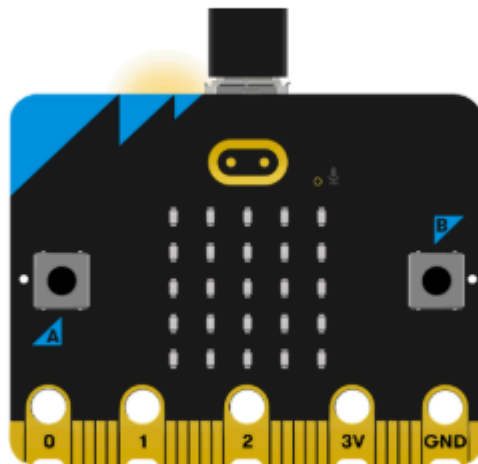
4. Program Download

1. Click the download button at the bottom left to download the program to the Micro:bit.



2. When the Micro USB data cable is not connected to the Microbit, the system will first prompt you to connect the device. If not connected, please connect the data cable to the Microbit first. If already connected, click [Next] here.

1. Connect your micro:bit to your computer



Next

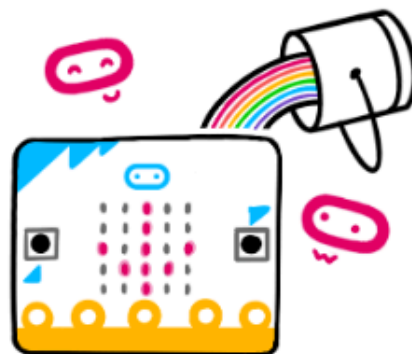
3. If the Microbit is detected as properly connected, the following interface will be displayed. At this point, you can click [Download].



Connected to micro:bit



Your micro:bit is connected! Pressing 'Download' will now automatically copy your code to your micro:bit.



Download

If the Microbit data cable is not properly connected or multiple Microbits are connected, the system will require you to manually select the device. You can choose between two methods: [Download as File] and [Pair].

2. Pair your micro:bit to your browser



Press the Pair button below.

A window will appear in the top of your browser.

Select the micro:bit device and click Connect.



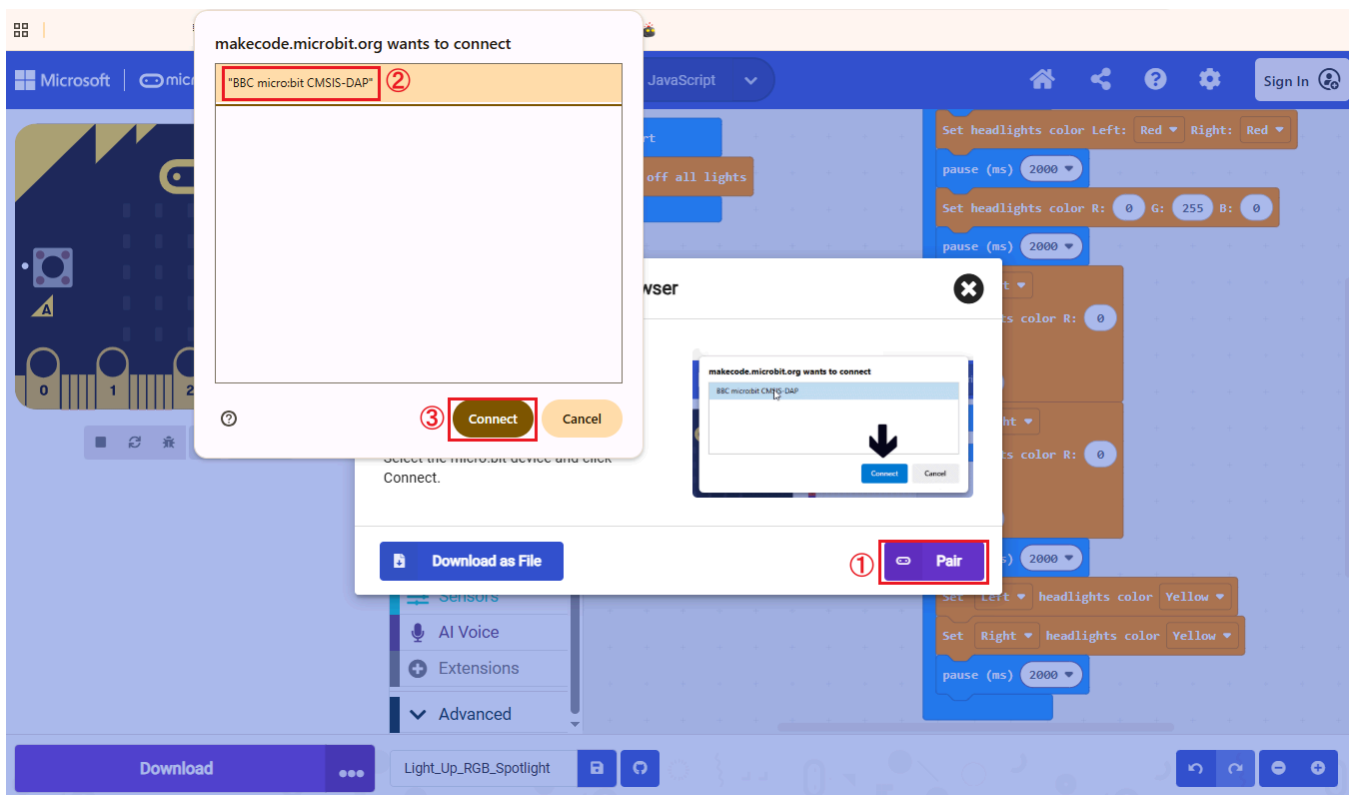
Download as File



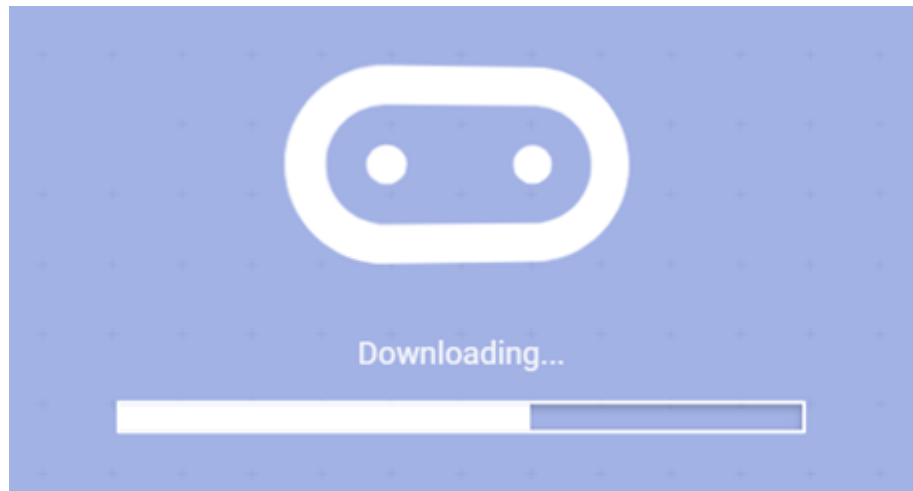
Pair

When selecting [Download as File], the system will download the currently opened program as `microbit-project-name.hex`. You can directly copy or drag this program to the corresponding MICROBIT drive letter to complete the download.

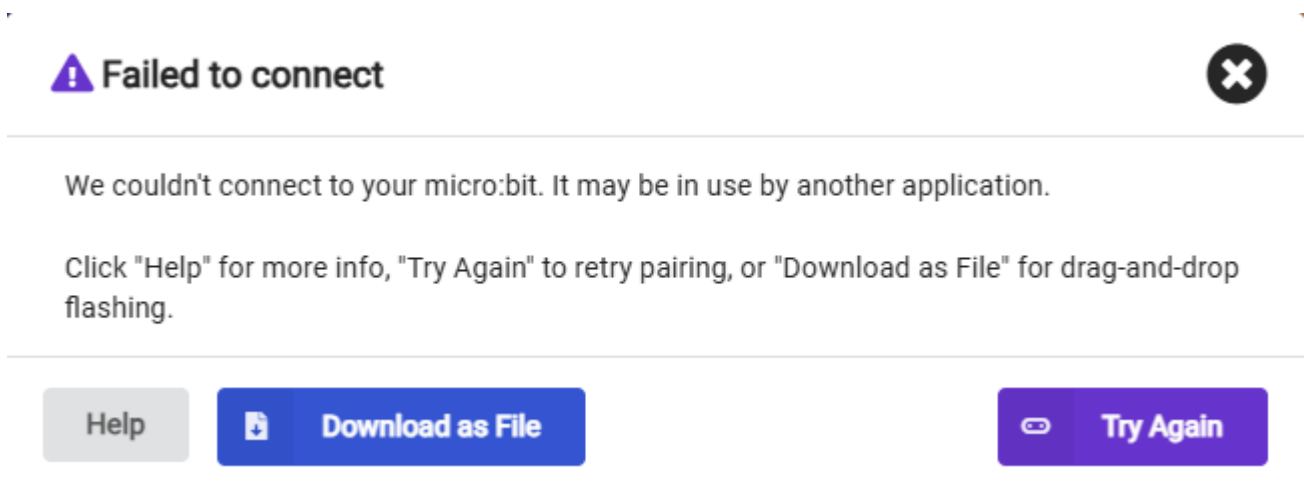
When selecting [Pair], you will be asked to manually select the device. Here, select `BBC micro:bit CMSIS:DAP`.



4. When the following interface appears, it means the program is being downloaded. Wait for the progress bar to complete to download the program to the Microbit.



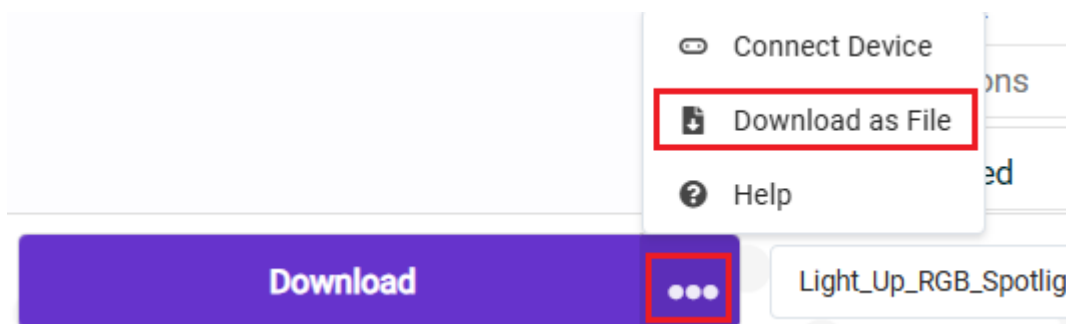
If multiple attempts continue to fail, please directly select [Download as File], save the current program as a file, and directly copy or drag the program to the corresponding MICROBIT drive letter to complete the download.



If you want to save the file directly, we provide two methods:

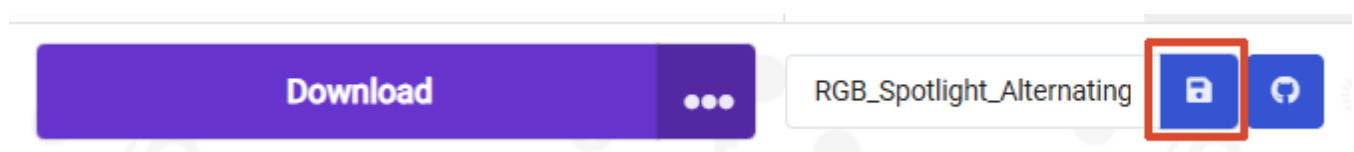
Method 1:

Click the [...] next to download, then select [Download as File]



Method 2:

Click [Save] after the [Project Name Edit Field].



5. Experiment Results

After flashing, turn on the switch (flip the switch to the ON position). The Micro:bit LED matrix screen will scroll to display the values from the photosensitive sensors, such as "Left:200 Right:200".